



YosysHQ

The Parable of the Square-Dancing Witches and Wizards

FMCAD 2026 - Sep 2026

Outline

- The three types of “sheep-and-goats” (SAG) bit-manipulation instructions
 - Why “Type (II) SAG” is the best of the three types. (Don Knuth agrees in TAOCP 4A.)
 - This type is sometime call “SAG with reversed goats”. Knuth calls the op gather-flip.
 - And the curious patent, that is in big part the reason, why no SAG insn is in RISC-V yet.
- The “witches-and-wizards” method for gather-flip (and its inverse: scatter-flip)
 - **Hands-on demo using the interactive tool at <https://clairexen.github.io/fmcad2026/>** ↘
 - The Parable of the Square-Dancing Witches and Wizards
 - Discussion of formally verified reference Verilog implementations
- Discussion of the Hilewitz patent and its claims
 - With focus of prior art for the “witches-and-wizards” method, esp. Fig. 4 of the 1994 paper M. J. Narasimha, “A Recursive Concentrator Structure with Applications to Self-Routing Switching Networks,” IEEE Transactions on Communications, 42(2–4), 896–898, 1994:
<https://doi.org/10.1109/TCOMM.1994.580197>.

⇒ <https://clairexen.github.io/fmcad2026/slides.pdf>



The Sheep and The Goat

The Sheep and the Goats

The bit-permutation operation that

- compresses the selected bits to the LSB end of the word,
- while the remaining bits are pushed to the MSB end [in reverse bit order],

is referred to as “sheep and goats” (SAG) operation [with reversed goat bits], in reference to The Parable of the Sheep and the Goats, found in Matthew 25:31-46.

Sources usually cite verses 32-33 of the parable when discussing the "sheep and goats" (SAG) operation and its unusual name:

All the nations will be gathered before him, and he will separate people one from another as a shepherd separates the sheep from the goats, and he will put the sheep at his right hand and the goats at the left.

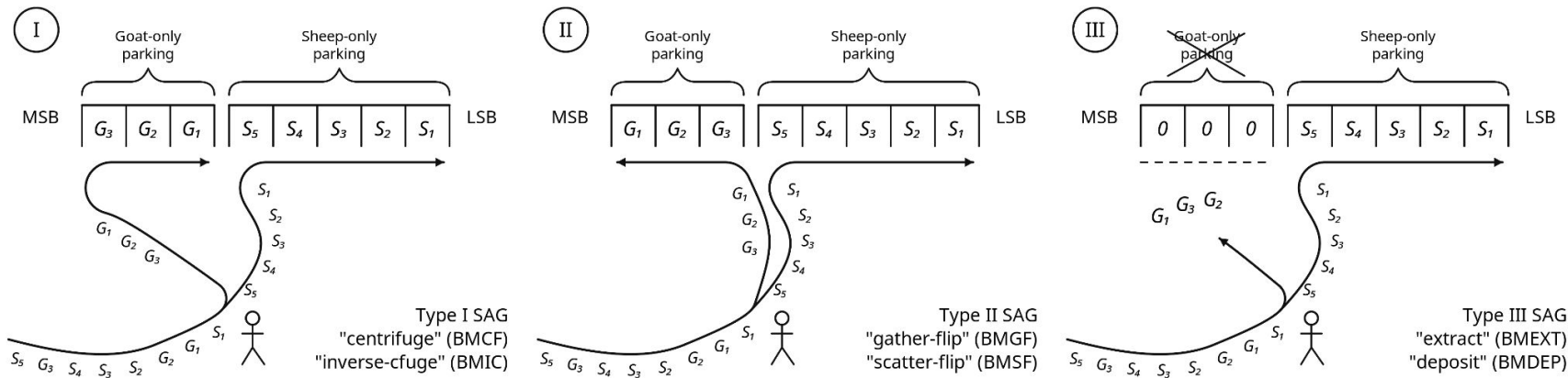
Sadly it's impossible to tell from this quote from scripture alone which order of goats bit is the “regular” ordering and which is the “reversed”, as the parable does not dive into the exact geometry of the situation.

The Sheep and the Goats

Further, verses 41-42 of the parable reveal a dire destiny for the goats:

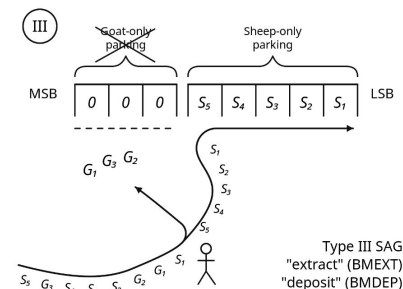
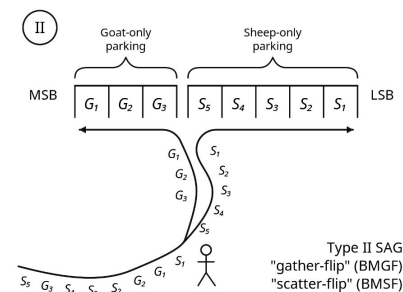
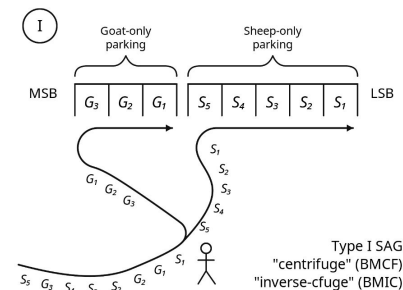
Then he will say to those on his left, 'Depart from me, you who are cursed, into the eternal fire prepared for the devil and his angels. For I was hungry and you gave me nothing to eat, I was thirsty and you gave me nothing to drink, [...]

This leaves us with three reasonable interpretations for the semantics of the “sheep-and-goats” bit-manipulation operation:



The Sheep and the Goats

- Type (I): the “centrifuge”
 - This is the most common interpretation
 - Knuth calls this operation the “sheep-and-goats” operation.
 - For example Power 10 and Cray X1/X1E have centrifuge CPU support
- Type (II): “gather-flip” and “scatter-flip”
 - Easier to implement than the centrifuge and arguably more powerful
 - Can emulate the other types easily but the others can’t easily emulate this type
 - However, I’m not aware of any commercially available CPU ISA, historic or present, with support for such an instruction.
 - US patents US8285766 and US9134953 seem to inadvertently cover the most efficient method I know for implementing these instructions.
 - That’s why I had to come up with an independent method.
 - Luckily I’ve found the circuit I’ve come up with independently was already published in 1994.
 - Thus, patents shouldn’t be an issue with this method.
- Type (III): “bit-extract” and “bit-deposit”
 - US patents US8285766 and US9134953 seem to cover this semantic
 - Implemented for example in X86 (PEXT/PDEP) and Power 10
 - The soviet BESM-6 mainframe computer, designed and built in the 1960s, had APX/AUX instructions with almost the same semantics: It packed/unpacked at the MSB end instead of the LSB end. Otherwise it is the same instruction.



Witches and Wizards

The “witches-and-wizards” method

- With $N=n^2$ signal bits, for each layer $i=0, 1, \dots, n-1$:
 - Split the control bit vector into 2^i partitions. Then for each partition:
 - Count the “unselected” (0) bits in the control bit vector, from LSB to MSB, and get the parity of that prefix-sum for each position. In other words: Calculate the XOR prefix sum of the inverted control bits.
 - For each pair of positions: swap the two control bits in a switch if the count of unselected bits to the right (LSB) of its center is odd.
 - Replicate this pattern of swaps with the bits in the data bit vector
 - Perform a “perfect unshuffle” on both control and data vector. That’s compressing all the even positions to the right (LSB) and all odd positions to the left (MSB). This corresponds to a rotate-right by 1 bit of the n -bit indices of the N positions. (Note that this is just wiring, no logic.)

Hands-on demo using an interactive JS widget

- <https://clairexen.github.io/fmcd2026/>
 - Demo and explain the witches and wizard method
 - Outline the Hilewitz and Lee method

The Parable of the Square-Dancing Witches and Wizards

- An island with $N=2^n$ witches and wizards, traveling/dancing from north towards south, in a long row extending in west-east direction.
 - The wizards, represented by a symbol for their magic wand (the digit 1), like the blue morning sky and are drawn towards to rising sun in the east.
 - The witches, represented by a symbol for their magic orbs (the digit 0), like the reddish-orange sky in the evening and are drawn towards to setting sun in the west.
 - In order to sort themselves according to these preferences they perform a 3-step square dance routine $n=\log_2(N)$ times:
- 1. Odd and even positions: Form pairs and hold hands. Look to your left (east) and see if the pair left of you is rising their hands (Condition 1). Then look if their right member is of the same gender as your left member (Condition 2). Now raise your hands when Condition 1 and Condition 2 are equal, that is when either both are satisfied or none is satisfied, and keep your hands down otherwise, when exactly one condition is satisfied. If you don't see a pair to your left, raise your hands when your left member is a witch and keep your hands down when your left member is a wizard.
- 2. If your hands are up even and odd are swapping places. Otherwise keep in your lane.
- 3. Perform a perfect un-shuffle: Even positions compress into the right half of the vector and odd positions to the left, while preserving order within the odd and even elements.

Formally Verified Verilog Reference Implementations

- <https://github.com/clairexen/fmcad2026>
 - Review bmgf.v and prove_bmgf.sby
 - Review hilewitz.v and bmext.v
 - Review stats_cached.md

The Hilewitz patent and its claims

<https://clairexen.github.io/fmcad2026/papers/hilewitz2006.pdf>

<https://clairexen.github.io/fmcad2026/papers/US8285766.pdf>

<https://clairexen.github.io/fmcad2026/papers/US9134953.pdf>

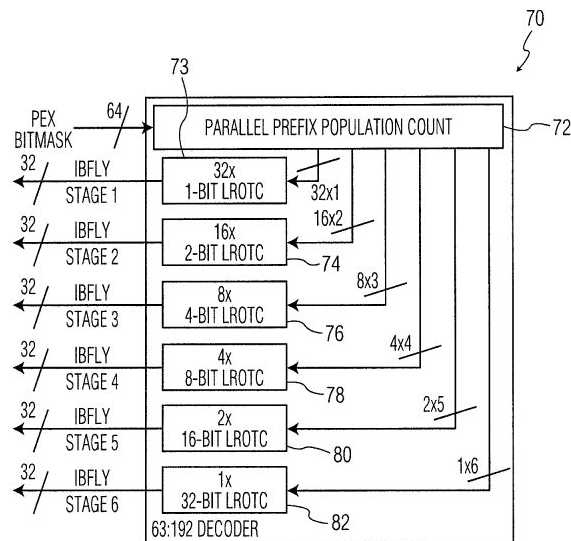
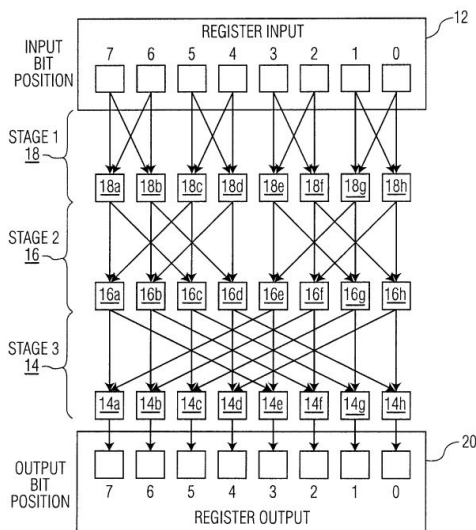
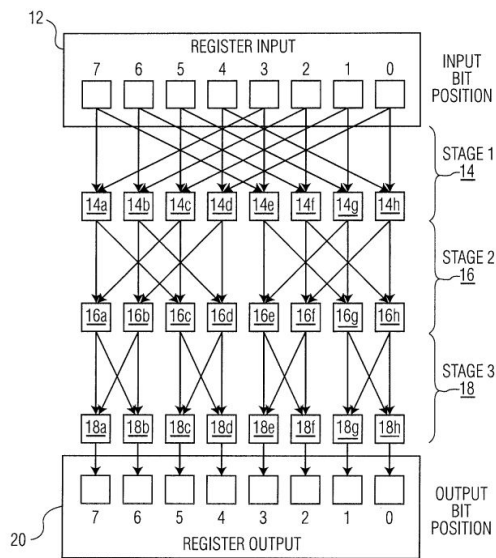


Fig. 4 in the Narasimha 1994 paper

<https://clairexen.github.io/fmcad2026/papers/narasimha1994.pdf>

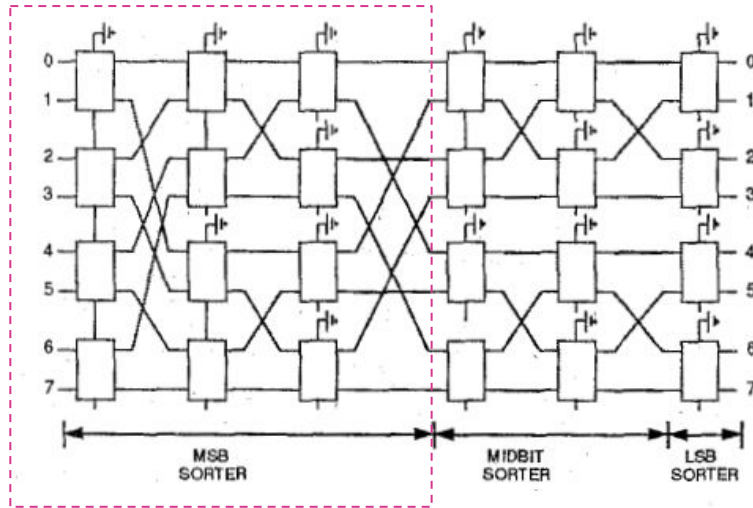


Fig. 6. An 8-input binary radix sorting network based on concentrators.

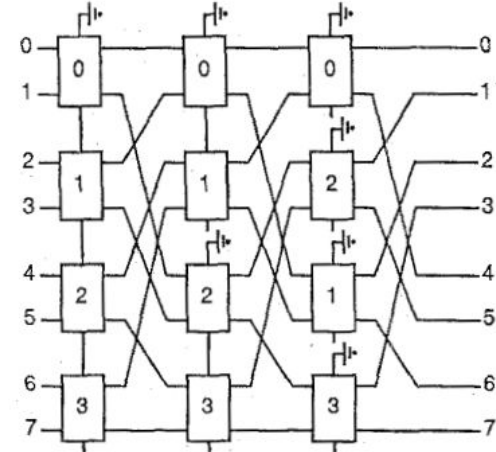


Fig. 5. The 8-input concentrator of Fig. 4 redrawn to demonstrate its topological similarity to the inverse omega network.

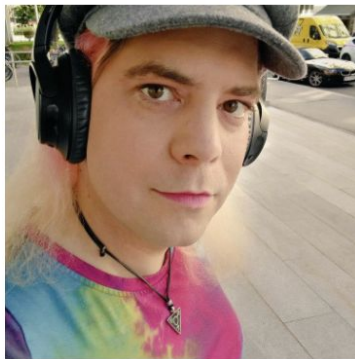
Thanks

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